

**NATIONAL UNIVERSITY OF MODERN LANGUAGE
ISLAMABAD**

Lab report



Submitted to:

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Subject: Operating System

Department: BSSE-4-Afternoon

Date: 05-06-2024

1: What is the file system used by the Linux?

The file system commonly used by Linux is called the Extended File System (ext). There have been several versions of the ext file system, with ext2, ext3, and ext4 being the most widely used. Each version has brought improvements in terms of performance, reliability, and features.

2: Name two types of boot loaders available.

Two types of boot loaders commonly used in Linux systems are:

GRUB (GRand Unified Bootloader): GRUB is one of the most popular boot loaders used in Linux distributions. It provides a flexible and feature-rich environment for booting multiple operating systems, configuring boot parameters, and troubleshooting boot-related issues. GRUB supports both BIOS-based (GRUB Legacy) and UEFI-based (GRUB 2) systems.

LILO (LIinux LOader): LILO was once a popular boot loader for Linux systems, but it has been largely replaced by GRUB. LILO is a simpler boot loader compared to GRUB and is primarily used in older systems or distributions that haven't adopted GRUB as the default boot loader. LILO has a configuration file named `lilo.conf` that defines the boot options and the location of the kernel image.

3: What are the names of partitions created for Linux?

Partitions created for Linux systems typically use a naming convention based on the device they are associated with. The most common naming scheme for partitions on Linux is based on the following pattern:

1. **Primary Partitions:** These are denoted by numbers (e.g., `/dev/sda1`, `/dev/sda2`, `/dev/sda3`, etc.). The first primary partition on the first disk is typically `/dev/sda1`, the second primary partition is `/dev/sda2`, and so on.
2. **Extended Partitions:** Historically, extended partitions were used to overcome the limit of four primary partitions per disk. Extended partitions do not directly contain file systems; instead, they serve as containers for logical partitions.
3. **Logical Partitions:** Logical partitions are partitions within extended partitions. They are also denoted by numbers but are typically numbered starting from 5 (e.g., `/dev/sda5`, `/dev/sda6`, `/dev/sda7`, etc.).

Lab No 02

task:- 01

```
zulqarnain@lenovo:~$ date
ت 09 مارچ 2024 و 16:30:41 PKT
zulqarnain@lenovo:~$ whoami
zulqarnain
zulqarnain@lenovo:~$ hostname
lenovo
zulqarnain@lenovo:~$ uname
Linux
zulqarnain@lenovo:~$ uptime
16:31:22 up 9 min, 1 user, load average: 0.43, 1.71, 1.26
zulqarnain@lenovo:~$
```

Task:- 02

```
zulqarnain@lenovo:~$ ls
Desktop Documents Downloads Music Pictures Public snap Templates Videos
zulqarnain@lenovo:~$ cat> hello.txt
Hello world!
this is a text file.
zulqarnain@lenovo:~$ ls
Desktop Documents Downloads hello.txt Music Pictures Public snap Templates Videos
zulqarnain@lenovo:~$ cat hello.txt
Hello world!
this is a text file.
zulqarnain@lenovo:~$ cat hello.txt
Hello world!
this is a text file.
```

Task:- 03

```
zulqarnain@lenovo:~$ cat> secret-of-the-univers
hi
I
am
a
BSse
student

zulqarnain@lenovo:~$ l
Desktop/   Downloads/ Music/    Public/   snap/     Videos/
Documents/ hello.txt Pictures/ secret-of-the-univers Templates/
zulqarnain@lenovo:~$ ls
Desktop Documents Downloads hello.txt Music Pictures Public secret-of-the-univers snap Templates Videos
zulqarnain@lenovo:~$ cat secret-of-the-univers
hi
I
am
a
BSse
student
```

Task:- 04

```
zulqarnain@lenovo:~$ cp secret-of-the-univers answer.txt
zulqarnain@lenovo:~$ cat answer.txt
hi
I
am
a
BSse
student

zulqarnain@lenovo:~$ cp hello.txt answer.txt
zulqarnain@lenovo:~$ cat answer.txt
Hello world
this is a text file.
zulqarnain@lenovo:~$ rm hello.txt
zulqarnain@lenovo:~$ ls
answer.txt Desktop Documents Downloads Music Pictures Public secret-of-the-univers snap Templates Videos
zulqarnain@lenovo:~$ mv answer.txt message
zulqarnain@lenovo:~$ ls
Desktop Documents Downloads message Music Pictures Public secret-of-the-univers snap Templates Videos
zulqarnain@lenovo:~$ rm missing
rm: cannot remove 'missing': No such file or directory
zulqarnain@lenovo:~$ cp secret-of-the-univers
cp: missing destination file operand after 'secret-of-the-univers'
Try 'cp --help' for more information.
zulqarnain@lenovo:~$
```

Lab No 03

task:- 01

```
zulqarnain@lenovo:/$ pwd
/
zulqarnain@lenovo:/$ cd /home/zulqarnain
zulqarnain@lenovo:~$ pwd
/home/zulqarnain
zulqarnain@lenovo:~$ cd ..
zulqarnain@lenovo:/home$ cd ..
zulqarnain@lenovo:/$ pwd
/
zulqarnain@lenovo:/$ ls
bin  boot  cdrom  dev  etc  home  lib  lib32  lib64  libx32  lost+found  media  mnt  opt  proc  root  run  sbin  snap  srv  swapfile  sys  tmp  usr  var
zulqarnain@lenovo:/$ cd home
zulqarnain@lenovo:/home$ ls
zulqarnain
zulqarnain@lenovo:/home$
```

task:- 02

```
zulqarnain@lenovo:~$ cat> shakespeare.txt
Shal i compare thee to a summer's day?
thou art more lovely and more temperate
zulqarnain@lenovo:~$ mv shakespeare.txt sonnet-18.txt
zulqarnain@lenovo:~$ mkdir poetry
zulqarnain@lenovo:~$ mv poem poetry
mv: cannot stat 'poem': No such file or directory
```

task:-3

```
zulqarnain@lenovo:/$ ls
bin boot cdrom dev etc home lib lib32 lib64 libx32 lost+found media mnt opt proc root run sbin snap srv swapfile sys tmp usr var
zulqarnain@lenovo:/$ cd usr/share
zulqarnain@lenovo:/usr/share$ ls
accountsservice dns gnome-remote-desktop keyrings open-vm-tools source-highlight
aclocal dnsmasq-base gnome-session language-selector openvpn speech-dispatcher
acpi-support doc gnome-settings-daemon language-support orca ssl-cert
adduser doc-base gnome-shell language-tools org.gnome.Characters system-config-printer
aisleriot dpkg gnome-system-monitor libc-bin os-prober systemd
alsa drirc.d gnome-todo libdebuginfod-common p11-kit tabset
alsa-base emacs gnome-video-effects libdrm PackageKit tcltk
alsa-card-profile emacsen-common gnupg libbexttextcat perl terminfo
appdata enchant-2 goa-1.0 libgnomekbd pam themes
applications eog grilo-plugins libgweather pam-configs thumbnails
apport evince grub libinput perl totem
apturl evolution-data-server grub2 liblangtag perl5 tracker3
aspell file grub-gfxpayload-lists liblouis pipewire transmission
avahi file-roller gst-plugins-base liblouisutdm1 pixmaps ubuntu
backgrounds fish gstreamer-1.0 libmysofa pkgconf ubuntu-drivers-common
base-files fontconfig gtk-4.0 libreoffice plymouth ubuntu-release-upgrader
base-passwd fonts gtk-doc librevenge pnm2ppa ubuntu-wayland
bash-completion fonts-droid-fallback gtk-engines libthai poppler ucf
binfmts fonts-abyssinica gtk-sourceview-2.0 lightdm polkit-1 unattended-upgrades
branding fonts-sil-padauk gtk-sourceview-3.0 lintian ppp unity
brltty foo2qpd1 guile gupnp-av locale ppp update-manager
bug foo2zjs gamemode gupnp-dlna-2.0 publicsuffix update-notifier
build-essential fwupd gcc gvfs lto-disabled-list upstart
ca-certificates gconf help help-langpack locales usb-creator
cnae GConf hunspell m2300w python3 usb_modeswitch
cogl gdb hplip man man-db python-apt vala
color colord gdm hunspell man-db python-apt vala
common-licenses gedit hunspell man-db python-apt vala
consolefonts gettext hunspell man-db python-apt vala
console-setup ghostscript i18n ibus maven-repo readline vulkan
consoletrans gir-1.0 i18n ibus media-player-info rhythmbox wayland-sessions
cups glade i18n ibus menu metainfo rsync xdg-desktop-portal
dbus-1 glib-2.0 icons mfx mime seahorse xml
dconf glvnd im-config mime secureboot xml-core
debconf gnome indicators misc session-migration xsessions
debhelper gnome-background-properties info mobile-broadband-provider-info sensible-utils yelp
debianutils gnome-bluetooth initramfs-tools ModemManager session-migration yelp-xsl
defaults gnome-bluetooth-3.0 installed-tests mouse tweaks sql zenity
```

```
zulqarnain@lenovo:/usr/share$ cd ..
zulqarnain@lenovo:/usr$ ls
bin games include lib lib32 lib64 libexec libx32 local sbin share src
zulqarnain@lenovo:/usr$ cd ..
zulqarnain@lenovo:/$ ls
bin boot cdrom dev etc home lib lib32 lib64 libx32 lost+found media mnt opt proc root run sbin snap srv swapfile sys tmp usr var
zulqarnain@lenovo:/$ echo ~
/home/zulqarnain
zulqarnain@lenovo:/$ cat ~/message /
Hello world
this is a text file.
cat: /: Is a directory
zulqarnain@lenovo:/$
```

task:- 04

```
zulqarnain@lenovo:/$ hostname
lenovo
zulqarnain@lenovo:/$ locate hostname
/etc/hostname
/snap/core22/1122/etc/hostname
/snap/core22/1122/etc/writable/hostname
/snap/core22/1122/usr/bin/hostname
/snap/core22/1122/usr/bin/hostnamectl
/snap/core22/1122/usr/lib/python3/dist-packages/cloudinit/config/cc_set_hostname.py
/snap/core22/1122/usr/lib/python3/dist-packages/cloudinit/config/cc_update_hostname.py
/snap/core22/1122/usr/lib/python3/dist-packages/cloudinit/config/_pycache_/cc_set_hostname.cpython-310.pyc
/snap/core22/1122/usr/lib/python3/dist-packages/cloudinit/config/_pycache_/cc_update_hostname.cpython-310.pyc
/snap/core22/1122/usr/lib/python3/dist-packages/cloudinit/distros/parsers/hostname.py
/snap/core22/1122/usr/lib/python3/dist-packages/cloudinit/distros/parsers/_pycache_/hostname.cpython-310.pyc
/snap/core22/1122/usr/lib/python3/dist-packages/urllib3/packages/ssl_match_hostname
/snap/core22/1122/usr/lib/python3/dist-packages/urllib3/packages/ssl_match_hostname/_init_.py
/snap/core22/1122/usr/lib/python3/dist-packages/urllib3/packages/ssl_match_hostname/_pycache_
/snap/core22/1122/usr/lib/python3/dist-packages/urllib3/packages/ssl_match_hostname/_implementation.py
/snap/core22/1122/usr/lib/python3/dist-packages/urllib3/packages/ssl_match_hostname/_pycache_/_init_.cpython-310.pyc
/snap/core22/1122/usr/lib/python3/dist-packages/urllib3/packages/ssl_match_hostname/_pycache_/_implementation.cpython-310.pyc
/snap/core22/1122/usr/lib/systemd/systemd-hostnamed
/snap/core22/1122/usr/lib/systemd/system/dbus-org.freedesktop.hostname1.service
/snap/core22/1122/usr/lib/systemd/system/systemd-hostnamed.service
/snap/core22/1122/usr/share/bash-completion/completions/hostname
/snap/core22/1122/usr/share/bash-completion/completions/hostnamectl
/snap/core22/1122/usr/share/dbus-1/system-services/org.freedesktop.hostname1.service
/snap/core22/1122/usr/share/dbus-1/system.d/org.freedesktop.hostname1.conf
/snap/core22/1122/usr/share/doc/hostname
/snap/core22/1122/usr/share/doc/hostname/changelog.gz
/snap/core22/1122/usr/share/doc/hostname/copyright
/snap/core22/1122/usr/share/polkit-1/actions/org.freedesktop.hostname1.policy
/snap/core22/1122/usr/share/zsh/vendor-completions/_hostnamectl
/snap/gnome-42-2204/141/usr/lib/python3/dist-packages/pip/_vendor/urllib3/util/ssl_match_hostname.py
/snap/gnome-42-2204/141/usr/lib/x86_64-linux-gnu/gettext/hostname
/usr/bin/hostname
/usr/bin/hostnamectl
/usr/bin/nmtui-hostname
/usr/lib/python3/dist-packages/urllib3/packages/ssl_match_hostname
/usr/lib/python3/dist-packages/urllib3/packages/ssl_match_hostname/_init_.py
/usr/lib/python3/dist-packages/urllib3/packages/ssl_match_hostname/_pycache_
/usr/lib/python3/dist-packages/urllib3/packages/ssl_match_hostname/_implementation.py
/usr/lib/python3/dist-packages/urllib3/packages/ssl_match_hostname/_pycache_/_init_.cpython-310.pyc
/usr/lib/python3/dist-packages/urllib3/packages/ssl_match_hostname/_pycache_/_implementation.cpython-310.pyc
/usr/lib/systemd/systemd-hostnamed
/usr/lib/systemd/system/dbus-org.freedesktop.hostname1.service
/usr/lib/systemd/system/systemd-hostnamed.service
/usr/share/bash-completion/completions/hostname
/usr/share/bash-completion/completions/hostnamectl
/usr/share/dbus-1/system-services/org.freedesktop.hostname1.service
/usr/share/dbus-1/system.d/org.freedesktop.hostname1.conf
/usr/share/doc/hostname
/usr/share/doc/hostname/changelog.gz
/usr/share/doc/hostname/copyright
/usr/share/man/man1/hostname.1.gz
/usr/share/man/man1/hostnamectl.1.gz
```

```
zulqarnain@lenovo:/$ cd user/bin/hostname
bash: cd: user/bin/hostname: No such file or directory
zulqarnain@lenovo:/$ cd /user/bin/hostname
bash: cd: /user/bin/hostname: No such file or directory
zulqarnain@lenovo:/$ cd /usr/bin/hostname
bash: cd: /usr/bin/hostname: Not a directory
zulqarnain@lenovo:/$ cd usr/bin/hostname
bash: cd: usr/bin/hostname: Not a directory
zulqarnain@lenovo:/$
```

task:- 05

```
zulqarnain@lenovo:~$ echo *
Desktop Documents Downloads message Music Pictures poetry Public secret-of-the-univers snap sonnet-18.txt Templates Videos
zulqarnain@lenovo:~$ echo '*'
*
zulqarnain@lenovo:~$ ls
Desktop Documents Downloads message Music Pictures poetry Public secret-of-the-univers snap sonnet-18.txt Templates Videos
zulqarnain@lenovo:~$ cp poeam.txt sonnet-18.txt
cp: cannot stat 'poeam.txt': No such file or directory
zulqarnain@lenovo:~$ cp sonnet-18.txt poeam.txt
zulqarnain@lenovo:~$ ls
Desktop Documents Downloads message Music Pictures poeam.txt poetry Public secret-of-the-univers snap sonnet-18.txt Templates Videos
zulqarnain@lenovo:~$ echo *
Desktop Documents Downloads message Music Pictures poeam.txt poetry Public secret-of-the-univers snap sonnet-18.txt Templates Videos
zulqarnain@lenovo:~$ echo '*'
*
zulqarnain@lenovo:~$ cat *.txt
Shal i compare thee to a summer's day?
thou art more lovely and more temperate
Shal i compare thee to a summer's day?
thou art more lovely and more temperate
zulqarnain@lenovo:~$ rmdir poetry
zulqarnain@lenovo:~$ ls
Desktop Documents Downloads message Music Pictures poeam.txt Public secret-of-the-univers snap sonnet-18.txt Templates Videos
zulqarnain@lenovo:~$
```


Lab task:-04

Task 1

```
zulqarnain@lenovo:~/Desktop$ df
Filesystem      1K-blocks    Used Available Use% Mounted on
tmpfs            507516      1624    505892   1% /run
/dev/sda3       25106692 12601228  11204780  53% /
tmpfs           2537580        0   2537580   0% /dev/shm
tmpfs            5120         4     5116    1% /run/lock
/dev/sda2       524252      6220    518032   2% /boot/efi
tmpfs           507516      112    507404   1% /run/user/1000
/dev/sr0        52272      52272         0 100% /media/zulqarnain/VBox_GAs_7.0.14
zulqarnain@lenovo:~/Desktop$ man df
zulqarnain@lenovo:~/Desktop$ bash
zulqarnain@lenovo:~/Desktop$ exit
exit
zulqarnain@lenovo:~/Desktop$
```

Task 2

```
zulqarnain@lenovo:~/Desktop$ ls -a
. . .
zulqarnain@lenovo:~/Desktop$ ls -A
zulqarnain@lenovo:~/Desktop$ for file in *; do echo "$file"; done
*
zulqarnain@lenovo:~/Desktop$ for person in Alice Bob Charlie David; do echo "Hello, $person !"; done
Hello, Alice !
Hello, Bob !
Hello, Charlie !
Hello, David !
zulqarnain@lenovo:~/Desktop$ echo *
*
zulqarnain@lenovo:~/Desktop$ ls
zulqarnain@lenovo:~/Desktop$
```

Task 3

```
zulqarnain@lenovo:/home$ find ~
/home/zulqarnain
/home/zulqarnain/tut1.cpp
/home/zulqarnain/.vboxclient-hostversion-tty2-control.pid
/home/zulqarnain/sonnet-18.txt
/home/zulqarnain/.vboxclient-clipboard-tty2-control.pid
/home/zulqarnain/message
/home/zulqarnain/.bashrc
/home/zulqarnain/.lessht
/home/zulqarnain/a.out
/home/zulqarnain/.config
/home/zulqarnain/.config/evolution
/home/zulqarnain/.config/evolution/sources
/home/zulqarnain/.config/evolution/sources/system-calendar.source
/home/zulqarnain/.config/evolution/sources/birthdays.source
/home/zulqarnain/.config/evolution/sources/system-proxy.source
/home/zulqarnain/.config/gtk-3.0
/home/zulqarnain/.config/gtk-3.0/bookmarks
/home/zulqarnain/.config/mimeapps.list
/home/zulqarnain/.config/gnome-initial-setup-done
/home/zulqarnain/.config/user-dirs.dirs
/home/zulqarnain/.config/.gsd-keyboard.settings-portcd
/home/zulqarnain/.config/pulse
/home/zulqarnain/.config/pulse/c90240c411d34b05be042a5fc91070a3-stream-volumes.tdb
/home/zulqarnain/.config/pulse/c90240c411d34b05be042a5fc91070a3-device-volumes.tdb
/home/zulqarnain/.config/pulse/c90240c411d34b05be042a5fc91070a3-card-database.tdb
/home/zulqarnain/.config/pulse/c90240c411d34b05be042a5fc91070a3-default-source
/home/zulqarnain/.config/pulse/cookie
/home/zulqarnain/.config/pulse/c90240c411d34b05be042a5fc91070a3-default-sink
/home/zulqarnain/.config/dconf
```

```
zulqarnain@lenovo:/home$ find ~ -type f
/home/zulqarnain/tut1.cpp
/home/zulqarnain/.vboxclient-hostversion-tty2-control.pid
/home/zulqarnain/sonnet-18.txt
/home/zulqarnain/.vboxclient-clipboard-tty2-control.pid
/home/zulqarnain/message
/home/zulqarnain/.bashrc
/home/zulqarnain/.lessht
/home/zulqarnain/a.out
/home/zulqarnain/.config/evolution/sources/system-calendar.source
/home/zulqarnain/.config/evolution/sources/birthdays.source
/home/zulqarnain/.config/evolution/sources/system-proxy.source
/home/zulqarnain/.config/gtk-3.0/bookmarks
/home/zulqarnain/.config/mimeapps.list
/home/zulqarnain/.config/gnome-initial-setup-done
/home/zulqarnain/.config/user-dirs.dirs
/home/zulqarnain/.config/.gsd-keyboard.settings-portcd
/home/zulqarnain/.config/pulse/c90240c411d34b05be042a5fc91070a3-stream-volumes.tdb
/home/zulqarnain/.config/pulse/c90240c411d34b05be042a5fc91070a3-device-volumes.tdb
/home/zulqarnain/.config/pulse/c90240c411d34b05be042a5fc91070a3-card-database.tdb
/home/zulqarnain/.config/pulse/c90240c411d34b05be042a5fc91070a3-default-source
/home/zulqarnain/.config/pulse/cookie
/home/zulqarnain/.config/pulse/c90240c411d34b05be042a5fc91070a3-default-sink
/home/zulqarnain/.config/dconf/user
/home/zulqarnain/.config/ibus/bus/c90240c411d34b05be042a5fc91070a3-unix-0
/home/zulqarnain/.config/ibus/bus/c90240c411d34b05be042a5fc91070a3-unix-wayland-0
/home/zulqarnain/.config/user-dirs.locale
/home/zulqarnain/.vboxclient/vmsvga-session-tty2-control.pid
```

```
zulqarnain@lenovo:/home$ find ~ -type d
/home/zulqarnain
/home/zulqarnain/.config
/home/zulqarnain/.config/evolution
/home/zulqarnain/.config/evolution/sources
/home/zulqarnain/.config/gtk-3.0
/home/zulqarnain/.config/pulse
/home/zulqarnain/.config/dconf
/home/zulqarnain/.config/gnome-session
/home/zulqarnain/.config/gnome-session/saved-session
/home/zulqarnain/.config/update-notifier
/home/zulqarnain/.config/ibus
/home/zulqarnain/.config/ibus/bus
/home/zulqarnain/.config/goa-1.0
/home/zulqarnain/.config/nautilus
/home/zulqarnain/Templates
/home/zulqarnain/.local
/home/zulqarnain/.local/share
/home/zulqarnain/.local/share/gnome-shell
/home/zulqarnain/.local/share/gnome-settings-daemon
/home/zulqarnain/.local/share/sounds
/home/zulqarnain/.local/share/evolution
/home/zulqarnain/.local/share/evolution/mail
/home/zulqarnain/.local/share/evolution/mail/trash
/home/zulqarnain/.local/share/evolution/tasks
/home/zulqarnain/.local/share/evolution/tasks/trash
/home/zulqarnain/.local/share/evolution/tasks/system
/home/zulqarnain/.local/share/evolution/calendar
/home/zulqarnain/.local/share/evolution/calendar/trash
/home/zulqarnain/.local/share/evolution/calendar/system
```

```
zulqarnain@lenovo:/home$ locate bashbug
/snap/core22/1122/usr/bin/bashbug
/usr/bin/bashbug
/usr/share/man/man1/bashbug.1.gz
zulqarnain@lenovo:/home$ sudo find / -type f -name '*bashbug*'
[sudo] password for zulqarnain:
find: '/run/user/1000/doc': Permission denied
find: '/run/user/1000/gvfs': Permission denied
/snap/core22/1122/usr/bin/bashbug
/usr/bin/bashbug
/usr/share/man/man1/bashbug.1.gz
zulqarnain@lenovo:/home$ find / -type f -iname '*bashbug*'
find: '/sys/kernel/tracing': Permission denied
find: '/sys/kernel/debug': Permission denied
find: '/sys/fs/pstore': Permission denied
find: '/sys/fs/bpf': Permission denied
find: '/etc/polkit-1/localauthority': Permission denied
find: '/etc/ssl/private': Permission denied
find: '/etc/cups/ssl': Permission denied
find: '/boot/efi': Permission denied
find: '/run/gdm3': Permission denied
find: '/run/udisks2': Permission denied
find: '/run/cups/certs': Permission denied
find: '/run/user/1000/systemd/inaccessible/dir': Permission denied
find: '/run/sudo': Permission denied
find: '/run/speech-dispatcher': Permission denied
find: '/run/openvpn-server': Permission denied
find: '/run/openvpn-client': Permission denied
```

Lab task:-05

Task #01

```
zulqarnain@lenovo:~$ mkdir dog
zulqarnain@lenovo:~$ cd dog
zulqarnain@lenovo:~/dog$ mkdir cat
zulqarnain@lenovo:~/dog$ mkdir cat/mouse
zulqarnain@lenovo:~/dog$ cd ..
zulqarnain@lenovo:~$ rm -r /dog
rm: cannot remove '/dog': No such file or directory
zulqarnain@lenovo:~$ ls
a.out      dog      Music     Public     sonnet-18.txt  Videos
Desktop    Downloads Pictures    secret-of-the-univers  Templates
Documents  message  poeam.txt  snap       tut1.cpp
zulqarnain@lenovo:~$ rmdir dog
rmdir: failed to remove 'dog': Directory not empty
zulqarnain@lenovo:~$ cd dog
zulqarnain@lenovo:~/dog$ rm -r ~/cat
rm: cannot remove '/home/zulqarnain/cat': No such file or directory
zulqarnain@lenovo:~/dog$ cd ..
zulqarnain@lenovo:~$ rm -r ~/dog
zulqarnain@lenovo:~$
```

Task #02

```
zulqarnain@lenovo:~/Desktop$ cp password ~/password_copy
zulqarnain@lenovo:~/Desktop$ cat ~/password_copy
a
b
c
d

zulqarnain@lenovo:~/Desktop$ mv ~/password_copy ~/users
zulqarnain@lenovo:~/Desktop$ cd ..
zulqarnain@lenovo:~$ mkdir programs
zulqarnain@lenovo:~$ cp -r /bin/* ~/programs/
zulqarnain@lenovo:~$ rm -rf ~/programs/*
zulqarnain@lenovo:~$ rm -rf ~/programs/
zulqarnain@lenovo:~$ rm ~/users
zulqarnain@lenovo:~$
```

Task #03

```
zulqarnain@lenovo:~$ touch baked-beans
zulqarnain@lenovo:~$ ls -l baked-beans
-rw-rw-r-- 1 zulqarnain zulqarnain 0 10:07 28   مارج baked-beans
zulqarnain@lenovo:~$ touch baked-beans
zulqarnain@lenovo:~$ ls -l baked-beans
-rw-rw-r-- 1 zulqarnain zulqarnain 0 10:08 28   مارج baked-beans
zulqarnain@lenovo:~$ touch -t 2024412312359.59 baked-beans
touch: invalid date format '2024412312359.59'
zulqarnain@lenovo:~$ touch -t 202041221235 baked-beans
touch: invalid date format '202041221235'
zulqarnain@lenovo:~$ touch -t 202041221235
touch: invalid date format '202041221235'
zulqarnain@lenovo:~$ rm baked-beans
zulqarnain@lenovo:~$
```

Lab task:-06

Task 01

```
zulqarnain@lenovo:~/Desktop$ who
zulqarnain tty2          2024-04-27 08:50 (tty2)
zulqarnain@lenovo:~/Desktop$ who | cut -d" " -f1
zulqarnain
zulqarnain@lenovo:~/Desktop$ who | sort | uniq -u
zulqarnain tty2          2024-04-27 08:50 (tty2)
zulqarnain@lenovo:~/Desktop$ last
zulqarna tty2            tty2            Sat Apr 27 08:50    still logged in
reboot  system boot      6.5.0-26-generic  Sat Apr 27 08:47    still running
zulqarna tty2            tty2            Mon Apr 22 10:29 - crash (4+22:18)
reboot  system boot      6.5.0-26-generic  Mon Apr 22 10:27    still running
zulqarna tty2            tty2            Thu Mar 28 09:38 - crash (25+00:48)
zulqarna tty2            tty2            Thu Mar 28 09:38 - 09:38 (00:00)
reboot  system boot      6.5.0-26-generic  Thu Mar 28 09:36    still running
zulqarna tty2            tty2            Sun Mar 10 21:39 - crash (17+11:57)
reboot  system boot      6.5.0-25-generic  Sun Mar 10 21:37    still running
zulqarna tty2            tty2            Sat Mar 9 14:01 - crash (1+07:35)
reboot  system boot      6.5.0-25-generic  Sat Mar 9 14:00    still running
zulqarna tty2            tty2            Sat Mar 9 13:43 - down (00:17)
reboot  system boot      6.5.0-25-generic  Sat Mar 9 13:42 - 14:00 (00:18)

wtmp begins Sat Mar 9 13:42:15 2024
zulqarnain@lenovo:~/Desktop$ last | tac
wtmp begins Sat Mar 9 13:42:15 2024

reboot  system boot      6.5.0-25-generic  Sat Mar 9 13:42 - 14:00 (00:18)
zulqarna tty2            tty2            Sat Mar 9 13:43 - down (00:17)
reboot  system boot      6.5.0-25-generic  Sat Mar 9 14:00    still running
zulqarna tty2            tty2            Sat Mar 9 14:01 - crash (1+07:35)
reboot  system boot      6.5.0-25-generic  Sun Mar 10 21:37    still running
zulqarna tty2            tty2            Sun Mar 10 21:39 - crash (17+11:57)
reboot  system boot      6.5.0-26-generic  Thu Mar 28 09:36    still running
zulqarna tty2            tty2            Thu Mar 28 09:38 - 09:38 (00:00)
zulqarna tty2            tty2            Thu Mar 28 09:38 - crash (25+00:48)
reboot  system boot      6.5.0-26-generic  Mon Apr 22 10:27    still running
zulqarna tty2            tty2            Mon Apr 22 10:29 - crash (4+22:18)
reboot  system boot      6.5.0-26-generic  Sat Apr 27 08:47    still running
zulqarna tty2            tty2            Sat Apr 27 08:50    still logged in
zulqarnain@lenovo:~/Desktop$
```

```
zulqarnain@lenovo:~/Desktop$ cat > spelling.sed
s/enviroment/environment/g
zulqarnain@lenovo:~/Desktop$ cat>text.txt
envitoment is our enviromment
zulqarnain@lenovo:~/Desktop$ sed -f spelling.sed text.txt > text_file.txt
zulqarnain@lenovo:~/Desktop$ cat text_file.txt
envitoment is our enviromment
zulqarnain@lenovo:~/Desktop$ sed -f spelling.sed text.txt > text_file.txt | nl
zulqarnain@lenovo:~/Desktop$ cat text_file.txt | nl
1 envitoment is our enviromment
zulqarnain@lenovo:~/Desktop$
```


Task 02

```
zulqarnain@lenovo:~$ touch g.txt
zulqarnain@lenovo:~$ echo "this is sixth line" >>g.txt
zulqarnain@lenovo:~$ tr 'ABCDEF' '012345' g.txt
tr: extra operand 'g.txt'
Try 'tr --help' for more information.
zulqarnain@lenovo:~$ cat g.txt
this is sixth line
zulqarnain@lenovo:~$ /bin/ls -f
tut1.cpp                .config                .bash_logout          .vboxclient-seamless-tty2-control.pid
..                      .vboxclient-vmsvga-session-tty2-control.pid .vboxclient-vmsvga-session-tty2-service.pid Music
sonnet-18.txt.old       Templates              Pictures              .vboxclient-seamless-tty2-service.pid
poem.txt.old            .local                .gnupg               .vboxclient-draganddrop-tty2-service.pid
.vboxclient-hostversion-tty2-control.pid Documents             .vboxclient-clipboard-tty2-service.pid Videos
.vboxclient-clipboard-tty2-control.pid secret-of-the-univers .ssh                  .
message                 .sudo_as_admin_successful .cache                Downloads
.bashrc                 snap                  .bash_history         Public
.lessshst               g.txt                 .profile
a.out                   .vboxclient-draganddrop-tty2-control.pid Desktop

zulqarnain@lenovo:~$ less -f /bin/ls
zulqarnain@lenovo:~$
```

```
zulqarnain@lenovo:~$ od -t x /bin/ls
00000000 042577 043114 000402 000001 000000 000000 000000 000000
00000020 000003 000076 000001 000000 065240 000000 000000 000000
00000040 000100 000000 000000 000000 012050 000002 000000 000000
00000060 000000 000000 000100 000070 000015 000100 000037 000036
00000100 000006 000000 000004 000000 000100 000000 000000 000000
00000120 000100 000000 000000 000000 000100 000000 000000 000000
00000140 001330 000000 000000 000000 001330 000000 000000 000000
00000160 000010 000000 000000 000000 000003 000000 000004 000000
00000200 001430 000000 000000 000000 001430 000000 000000 000000
00000220 001430 000000 000000 000000 000034 000000 000000 000000
00000240 000034 000000 000000 000000 000001 000000 000000 000000
00000260 000001 000000 000004 000000 000000 000000 000000 000000
00000300 000000 000000 000000 000000 000000 000000 000000 000000
00000320 032130 000000 000000 000000 032130 000000 000000 000000
00000340 010000 000000 000000 000000 000000 000001 000000 000005
00000360 040000 000000 000000 000000 040000 000000 000000 000000
00000400 040000 000000 000000 000000 030221 000001 000000 000000
00000420 030221 000001 000000 000000 010000 000000 000000 000000
00000440 000001 000000 000004 000000 100000 000001 000000 000000
00000460 100000 000001 000000 000000 100000 000001 000000 000000
00000500 072130 000000 000000 000000 072130 000000 000000 000000
00000520 010000 000000 000000 000000 000001 000000 000006 000000
00000540 177720 000001 000000 000000 007720 000002 000000 000000
00000560 007720 000002 000000 000000 011250 000000 000000 000000
00000600 022560 000000 000000 000000 010000 000000 000000 000000
00000620 000002 000000 000006 000000 005130 000002 000000 000000
00000640 015130 000002 000000 000000 015130 000002 000000 000000
00000660 001000 000000 000000 000000 001000 000000 000000 000000
00000700 000010 000000 000000 000000 000004 000000 000004 000000
00000720 001470 000000 000000 000000 001470 000000 000000 000000
00000740 001470 000000 000000 000000 000060 000000 000000 000000
00000760 000060 000000 000000 000000 000010 000000 000000 000000
00010000 000004 000000 000000 000004 000000 001550 000000 000000
00010020 001550 000000 000000 000000 001550 000000 000000 000000
00010040 000104 000000 000000 000000 000104 000000 000000 000000
```

```
zulqarnain@lenovo:~$ od -t x1 /bin/ls
00000000 7f 45 4c 46 02 01 01 00 00 00 00 00 00 00 00
00000020 03 00 3e 00 01 00 00 00 a0 6a 00 00 00 00 00
00000040 40 00 00 00 00 00 00 00 28 14 02 00 00 00 00
00000060 00 00 00 00 40 00 38 00 0d 00 40 00 1f 00 1e
00000100 06 00 00 00 04 00 00 00 40 00 00 00 00 00 00
00000120 40 00 00 00 00 00 00 00 40 00 00 00 00 00 00
00000140 d8 02 00 00 00 00 00 00 d8 02 00 00 00 00 00
00000160 08 00 00 00 00 00 00 00 03 00 00 00 04 00 00
00000200 18 03 00 00 00 00 00 00 18 03 00 00 00 00 00
00000220 18 03 00 00 00 00 00 00 1c 00 00 00 00 00 00
00000240 1c 00 00 00 00 00 00 00 01 00 00 00 00 00 00
00000260 01 00 00 00 04 00 00 00 00 00 00 00 00 00 00
00000300 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00000320 58 34 00 00 00 00 00 00 58 34 00 00 00 00 00
00000340 00 10 00 00 00 00 00 00 01 00 00 00 05 00 00
00000360 00 40 00 00 00 00 00 00 00 40 00 00 00 00 00
00000400 00 40 00 00 00 00 00 00 91 30 01 00 00 00 00
00000420 91 30 01 00 00 00 00 00 00 10 00 00 00 00 00
00000440 01 00 00 00 04 00 00 00 00 80 01 00 00 00 00
00000460 00 80 01 00 00 00 00 00 00 80 01 00 00 00 00
00000500 58 74 00 00 00 00 00 00 58 74 00 00 00 00 00
00000520 00 10 00 00 00 00 00 00 01 00 00 00 06 00 00
00000540 d0 ff 01 00 00 00 00 00 d0 0f 02 00 00 00 00
00000560 d0 0f 02 00 00 00 00 00 a8 12 00 00 00 00 00
00000600 70 25 00 00 00 00 00 00 00 10 00 00 00 00 00
00000620 02 00 00 00 06 00 00 00 58 0a 02 00 00 00 00
00000640 58 1a 02 00 00 00 00 00 58 1a 02 00 00 00 00
00000660 00 02 00 00 00 00 00 00 00 02 00 00 00 00 00
00000700 08 00 00 00 00 00 00 00 04 00 00 00 04 00 00
00000720 38 03 00 00 00 00 00 00 38 03 00 00 00 00 00
00000740 38 03 00 00 00 00 00 00 30 00 00 00 00 00 00
00000760 30 00 00 00 00 00 00 00 08 00 00 00 00 00 00
00010000 04 00 00 00 04 00 00 00 68 03 00 00 00 00 00
00010020 68 03 00 00 00 00 00 00 68 03 00 00 00 00 00
00010040 44 00 00 00 00 00 00 00 44 00 00 00 00 00 00
00010060 04 00 00 00 00 00 00 00 53 e5 74 64 04 00 00
00010100 38 03 00 00 00 00 00 00 38 03 00 00 00 00 00
00010120 38 03 00 00 00 00 00 00 30 00 00 00 00 00 00
00010140 30 00 00 00 00 00 00 00 08 00 00 00 00 00 00
00010160 50 e5 74 64 04 00 00 00 cc cd 01 00 00 00 00
00010200 cc cd 01 00 00 00 00 00 cc cd 01 00 00 00 00
00010220 6c 05 00 00 00 00 00 00 6c 05 00 00 00 00 00
00010240 04 00 00 00 00 00 00 00 51 e5 74 64 06 00 00
```

Task 03

```
zulqarnain@lenovo:~$ mkdir chunkD
zulqarnain@lenovo:~$ cd chunkD
zulqarnain@lenovo:~/chunkD$
zulqarnain@lenovo:~/chunkD$ touch chunks
zulqarnain@lenovo:~/chunkD$ cd ..
zulqarnain@lenovo:~$ less -f /bin/ls >>chunkD/chunks
zulqarnain@lenovo:~$ cd chunkD
zulqarnain@lenovo:~/chunkD$ ls
chunks
zulqarnain@lenovo:~/chunkD$ split -b 1k chunks chunk
zulqarnain@lenovo:~/chunkD$ ls
chunkaa chunkan chunkba chunkbn chunkca chunkcn chunkda chunkdn chunkea chunken chunkfa
chunkab chunkao chunkbb chunkbo chunkcb chunkco chunkdb chunkdo chunkeb chunkeo chunkfb
chunkac chunkap chunkbc chunkbp chunkcc chunkcp chunkdc chunkdp chunkec chunkep chunkfc
chunkad chunkaq chunkbd chunkbq chunkcd chunkcq chunkdd chunkdq chunked chunkeq chunkfd
chunkae chunkar chunkbe chunkbr chunkce chunkcr chunkde chunkdr chunkee chunker chunkfe
chunkaf chunkas chunkbf chunkbs chunkcf chunkcs chunkdf chunkds chunkef chunkes chunkfs
chunkag chunkat chunkbg chunkbt chunkcg chunkct chunkdg chunkdt chunkeg chunket
chunkah chunkau chunkbh chunkbu chunkch chunkcu chunkdh chunkdu chunkeh chunkeu
chunkai chunkav chunkbi chunkbv chunkci chunkcv chunkdi chunkdv chunkei chunkev
chunkaj chunkaw chunkbj chunkbw chunkcj chunkcw chunkdj chunkdw chunkej chunkew
chunkak chunkax chunkbk chunkbx chunkck chunkcx chunkdk chunkdx chunkek chunkex
chunkal chunkay chunkbl chunkby chunkcl chunkcy chunkdl chunkdy chunkel chunkey
chunkam chunkaz chunkbn chunkbz chunkcm chunkcz chunkdn chunkdz chunkem chunkez
zulqarnain@lenovo:~/chunkD$
```

Lab task:-07

Task 01

```
zulqarnain@lenovo:~/Desktop$ echo 'hello world' | rev
dlrow olleh
zulqarnain@lenovo:~/Desktop$ whoami | rev
nlanraqulz
zulqarnain@lenovo:~/Desktop$ echo 'hello world' | cat
hello world
zulqarnain@lenovo:~/Desktop$
```

Task 02

```
zulqarnain@lenovo:~/Desktop$ ls --color
aaaa aaab aaac a.txt b.txt clear c.txt d.txt e.txt f.txt g.txt lab1 spelling.sed text_file.txt text.txt
zulqarnain@lenovo:~/Desktop$ ls --color | cat
aaaa
aaab
aaac
a.txt
b.txt
clear
c.txt
d.txt
e.txt
f.txt
g.txt
lab1
spelling.sed
text_file.txt
text.txt
zulqarnain@lenovo:~/Desktop$ ls --color | less
zulqarnain@lenovo:~/Desktop$
```

Lab task:-08

Task 1

```
zulqarnain@lenovo: ~/Desktop
zulqarnain@lenovo:~/Desktop$ grep 'http' /etc/services
# Updated from https://www.iana.org/assignments/service-names-port-numbers/service-names-port-numbers.xhtml .
http      80/tcp      www         # WorldWideWeb HTTP
https     443/tcp     # https protocol over TLS/SSL
https     443/udp     # HTTP/3
http-alt  8080/tcp    webcache    # WWW caching service
zulqarnain@lenovo:~/Desktop$
```

```
zulqarnain@lenovo: ~/Desktop
zulqarnain@lenovo:~/Desktop$ grep -v '#' /etc/services

echo          7/tcp
echo          7/udp
discard       9/tcp          sink null
discard       9/udp          sink null
sysstat       11/tcp         users
daytime       13/tcp
daytime       13/udp
netstat       15/tcp
gotd          17/tcp          quote
chargen       19/tcp         ttytst source
chargen       19/udp         ttytst source
ftp-data      20/tcp
ftp           21/tcp
fsp           21/udp          fspfd
telnet        23/tcp
smtp          25/tcp          mail
time          37/tcp         timserver
time          37/udp         timserver
whois         43/tcp         nicname
tacacs        49/udp
domain        53/udp
bootps        67/udp
bootpc        68/udp
tftp          69/udp
finger        79/tcp
sunrpc        111/udp         portmapper
auth          113/tcp         authentication tap ident
snmp          161/udp
snmp-trap     162/udp         snmptrap
cmip-man      163/udp
cmip-agent    164/tcp
cmip-agent    164/udp
ntp-event     319/udp
ntp-general   320/udp
rpc2portmap   369/tcp
codaaauth2    370/tcp
clearcase     371/udp         Clearcase
ldap          389/udp
svrloc        427/udp
kpasswd       464/tcp
kpasswd       464/udp
rtsp          554/udp
gmpq          628/tcp
```

```
zulqarnain@lenovo: ~/Desktop
zulqarnain@lenovo:~/Desktop$ grep -v '^#' /etc/services |grep -v '^$' /etc/services
# Network services, Internet style
#
# Updated from https://www.iana.org/assignments/service-names-port-numbers/service-names-port-numbers.xhtml .
#
# New ports will be added on request if they have been officially assigned
# by IANA and used in the real-world or are needed by a debian package.
# If you need a huge list of used numbers please install the nmap package.
# TCP port service multiplexer
tcpmux        1/tcp
echo          7/tcp
echo          7/udp
discard       9/tcp          sink null
discard       9/udp          sink null
sysstat       11/tcp         users
daytime       13/tcp
daytime       13/udp
netstat       15/tcp
gotd          17/tcp          quote
chargen       19/tcp         ttytst source
chargen       19/udp         ttytst source
ftp-data      20/tcp
ftp           21/tcp
fsp           21/udp          fspfd
ssh           22/tcp          # SSH Remote Login Protocol
telnet        23/tcp
smtp          25/tcp          mail
time          37/tcp         timserver
time          37/udp         timserver
whois         43/tcp         nicname
tacacs        49/tcp          # Login Host Protocol (TACACS)
tacacs        49/udp
domain        53/tcp          # Domain Name Server
domain        53/udp
bootps        67/udp
bootpc        68/udp
tftp          69/udp
gopher        70/tcp
finger        79/tcp
http          80/tcp          www # WorldWideWeb HTTP
kerberos      88/tcp         kerberos5 krb5 kerberos-sec # Kerberos v5
kerberos      88/udp         kerberos5 krb5 kerberos-sec # Kerberos v5
lso-tsap      102/tcp         tsap # part of ISODE
acr-nema      104/tcp         dicom # Digital Imag. & Comm. 300
pop3          110/tcp         pop-3 # POP version 3
sunrpc        111/tcp         portmapper # RPC 4.0 portnapper
```

```
zulqarnain@lenovo:~/Desktop$ grep -v '^#' /etc/services | grep -v '^$' /etc/services | sed 's|/.*||g'
```

```
# Network services, Internet style
```

```
#
```

```
# Updated from https:
```

```
#
```

```
# New ports will be added on request if they have been officially assigned
```

```
# by IANA and used in the real-world or are needed by a debian package.
```

```
# If you need a huge list of used numbers please install the nmap package.
```

tcpmux	1
echo	7
echo	7
discard	9
discard	9
systat	11
daytime	13
daytime	13
netstat	15
qotd	17
chargen	19
chargen	19
ftp-data	20
ftp	21
fsp	21
ssh	22
telnet	23
smtp	25
time	37
time	37
whois	43
tacacs	49
tacacs	49
domain	53
domain	53
bootps	67
bootpc	68
tftp	69
gopher	70
finger	79
http	80
kerberos	88
kerberos	88
iso-tsap	102
acr-nema	104
pop3	110
sunrpc	111

Lab task:-09

```
zulqarnain@lenovo:~/Desktop$ man bash &
[15] 4191
zulqarnain@lenovo:~/Desktop$ jobs
[1]  Stopped                  grep --color=auto -v '^#' /etc/services | less
[2]  Stopped                  grep --color=auto -v '^#' /etc/services | less
[3]  Stopped                  grep --color=auto -v '^#' /etc/services | less
[4]  Stopped                  grep --color=auto -v '^#' /etc/services | grep --color=auto -v '^$' /etc/services | less
[5]  Stopped                  grep --color=auto -v '^#' /etc/services | grep --color=auto -v '^$' /etc/services | sed '||.*||' | less
[6]  Stopped                  grep --color=auto -v '^#' /etc/services | grep --color=auto -v '^$' /etc/services | sed '||.*||' | less
[7]  Stopped                  man bash
[8]  Stopped                  man bash
[14]- Stopped                man bash
[15]+ Stopped                man bash
zulqarnain@lenovo:~/Desktop$ fg % 15
man bash
zulqarnain@lenovo:~/Desktop$ fg % 15
man bash

[14]+  Stopped                man bash
zulqarnain@lenovo:~/Desktop$ man bash

[15]+  Stopped                man bash
zulqarnain@lenovo:~/Desktop$ bg % 15
[15]+ man bash &
bash: bg: job 15 already in background
zulqarnain@lenovo:~/Desktop$

[15]+  Stopped                man bash
zulqarnain@lenovo:~/Desktop$
```